Hugo Marquerie 30/09/2025

Lem aditividad imp homogeneidad racionales

Lema 1. *Sea X un espacio vectorial sobre $\mathbb{K} = \mathbb{R}$ o $\mathbb{C}$ y $f\colon X \to \mathbb{K}$ aditiva*

$$\implies f \text{ es homogénea para escalares racionales } \lambda \in \mathbb{Q}.$$

***Demostración*:** Sea $x \in X$ y $\lambda \in \mathbb{Q}$.

Caso I ($\lambda \in \mathbb{N}$): Por aditividad de f, $f(nx) = f(x + \cdots + x) = f(x) + \cdots + f(x) = nf(x)$.

Caso II ($\lambda \in \mathbb{Z} \setminus \mathbb{N}$): Tenemos que $f(0) = f(0+0) = f(0) + f(0) \implies f(0) = 0$. Por tanto, $f(-nx) = f(0 - nx) = f(0) - f(nx) = -nf(x)$ por el Caso **??**.

Caso III ($\lambda \in \mathbb{Q}$): Sea $\lambda = \frac{p}{q}$ con $p \in \mathbb{Z}$, $q \in \mathbb{N}$. Entonces:

$$pf(x) \overset{\text{Caso ??}}{\overset{\downarrow}{=}} f(px) = f\left(q \cdot \frac{p}{q}x\right) \overset{\text{Caso ??}}{\overset{\downarrow}{=}} qf\left(\frac{p}{q}x\right) \implies f\left(\frac{p}{q}x\right) = \frac{p}{q}f(x).$$

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Referenciado en

- `lem-aditividad-continuidad-imp-homogeneidad`
- `lem-aditiva-no-homogenea-imp-grafica-densa`

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